
Reaching Agreements

An Introduction to MultiAgent Systems
<http://www.csc.liv.ac.uk/~mjw/pubs/imas>

Distributed Artificial Intelligence (DAI)

- Distributed Problem Solving (DPS)
 - Centrally designed systems, built-in cooperation, have *global* problem to solve
- ✓ Multi-Agent Systems (MAS)
 - ✓ Group of utility-maximizing heterogeneous agents co-existing in same environment, possibly competitive

Reaching Agreements

- How do agents *reaching agreements* when they are self interested?
- In an extreme case (zero sum encounter) no agreement is possible — but in most scenarios, there is potential for *mutually beneficial agreement* on matters of common interest
- The capabilities of *negotiation* and *argumentation* are central to the ability of an agent to reach such agreements

Heterogeneous, Self-motivated Agents

The systems:

- are not centrally designed
- do not have a notion of global utility
- are dynamic (e.g., new types of agents)
- will not act “benevolently” unless it is in their interest to do so

Machines Controlling and Sharing Resources

- Electrical grids (load balancing)
- Telecommunications networks (routing)
- PDA's (schedulers)
- Shared databases (intelligent access)
- Traffic control (coordination)

Broad Working Assumption

- Designers (from different companies, countries, etc.) come together to agree on *standards* for how their automated agents will interact (in a given domain)
- Discuss various possibilities and their tradeoffs, and agree on protocols, strategies, and social laws to be implemented in their machines

Mechanisms, Protocols, and Strategies

- Negotiation is governed by a particular *mechanism*, or *protocol*
- The mechanism defines the “rules of encounter” between agents
- *Mechanism design* is designing mechanisms so that they have certain desirable properties
- Given a particular protocol, how can a particular *strategy* be designed that individual agents can use?

Mechanism Design

- Desirable properties of mechanisms:
 - *Convergence/guaranteed success*
 - *Maximizing social welfare*
 - *Pareto efficiency*
 - *Individual rationality*
 - *Stability*
 - *Simplicity*
 - *Distribution*

The Aim of the Research

- Social engineering for communities of machines
 - The creation of interaction environments that foster certain kinds of social behavior

**The exploitation of game theory
tools for high-level protocol design**

Attributes of Standards

- ✓ *Efficient*: Pareto Optimal
- ✓ *Stable*: No incentive to deviate
- ✓ *Simple*: Low computational and communication cost
- ✓ *Distributed*: No central decision-maker
- ✓ *Symmetric*: Agents play equivalent roles

Designing protocols for specific classes of domains that satisfy some or all of these attributes

Phone Call Competition Example

- Customer wishes to place long-distance call
- Carriers simultaneously bid, sending proposed prices
- Phone automatically chooses the carrier (dynamically)

MCI

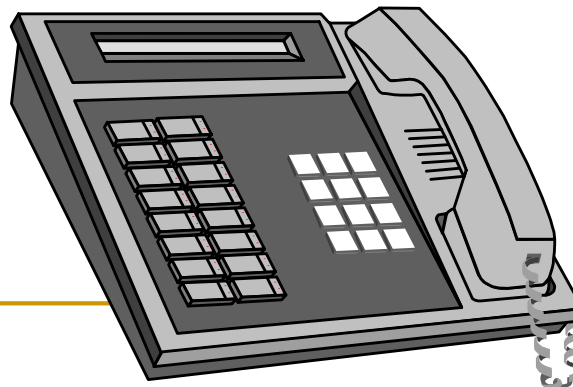
\$0.18

AT&T

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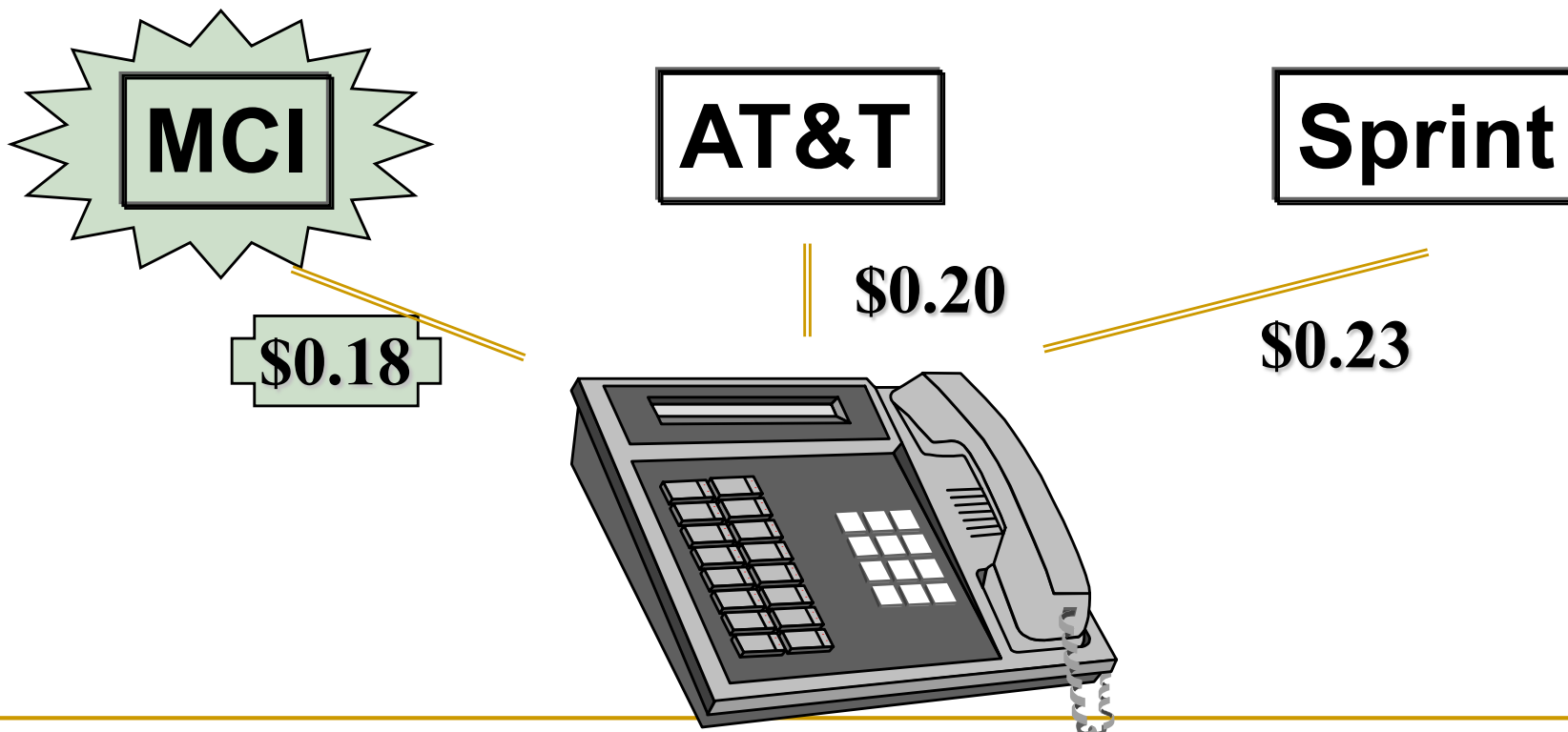
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Best Bid Wins

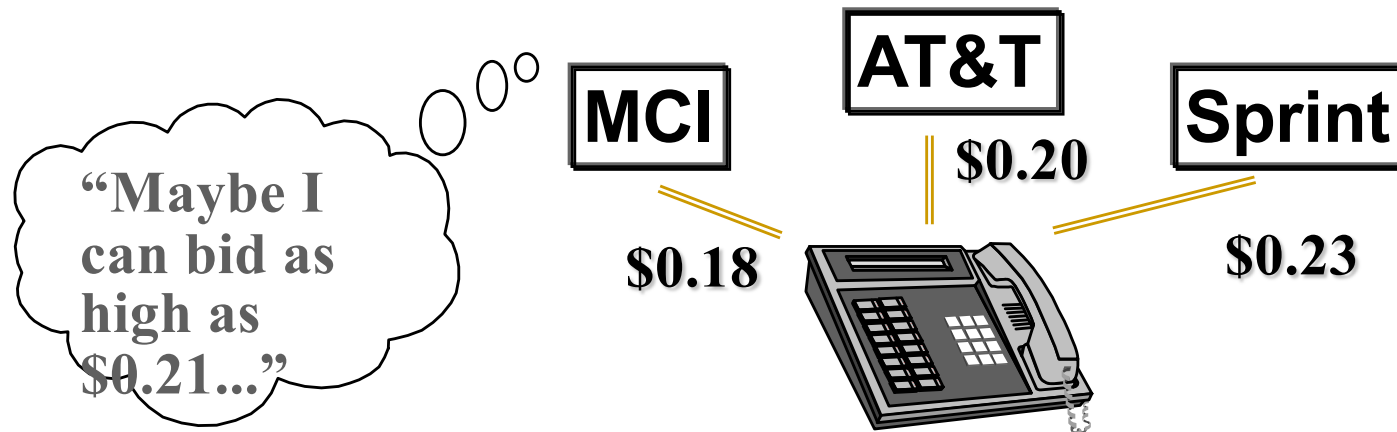
- Phone chooses carrier with lowest bid
- Carrier gets amount that it bid



Attributes of the Mechanism

- ✓ *Distributed*
- ✓ *Symmetric*
- ✗ *Stable*
- ✗ *Simple*
- ✗ *Efficient*

Carriers have an incentive to invest effort in strategic behavior



Auctions

- An auction takes place between an agent known as the *auctioneer* and a collection of agents known as the *bidders*
- The goal of the auction is for the auctioneer to allocate the *good* to one of the bidders
- In most settings the auctioneer desires to maximize the price; bidders desire to minimize price

Auction Parameters

- *Goods can have*
 - *private value*
 - *public/common value*
 - *correlated value*
- *Winner determination may be*
 - *first price*
 - *second price*
- *Bids may be*
 - *open cry*
 - *sealed bid*
- *Bidding may be*
 - *one shot*
 - *ascending*
 - *descending*

English Auctions

- Most commonly known type of auction:
 - *first price*
 - *open cry*
 - *ascending*
- Dominant strategy is for agent to successively bid a small amount more than the current highest bid until it reaches their valuation, then withdraw
- Susceptible to:
 - *winner's curse*
 - *shills*

Dutch Auctions

- Dutch auctions are examples of *open-cry descending* auctions:
 - ❑ auctioneer starts by offering good at artificially high value
 - ❑ auctioneer lowers offer price until some agent makes a bid equal to the current offer price
 - ❑ the good is then allocated to the agent that made the offer

First-Price Sealed-Bid Auctions

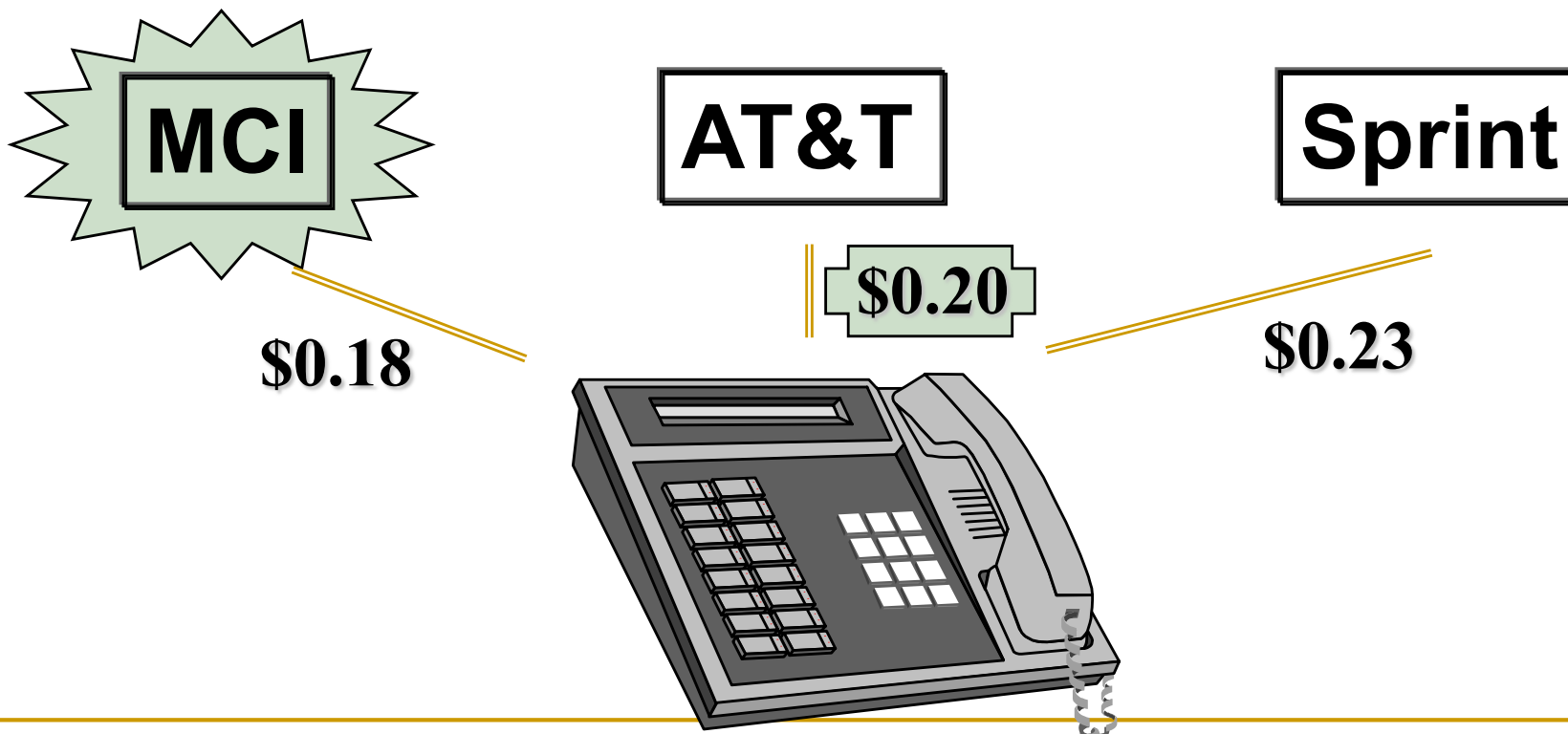
- First-price sealed-bid auctions are *one-shot auctions*:
 - there is a single round
 - bidders submit a sealed bid for the good
 - good is allocated to agent that made highest bid
 - winner pays price of highest bid
- Best strategy is to *bid less than true valuation*

Vickrey Auctions

- Vickrey auctions are:
 - *second-price*
 - *sealed-bid*
- Good is awarded to the agent that made the highest bid; at the price of the *second highest* bid
- *Bidding to your true valuation is dominant strategy in Vickrey auctions*
- Vickrey auctions susceptible to *antisocial* behavior

Best Bid Wins, Gets Second Price (Vickrey Auction)

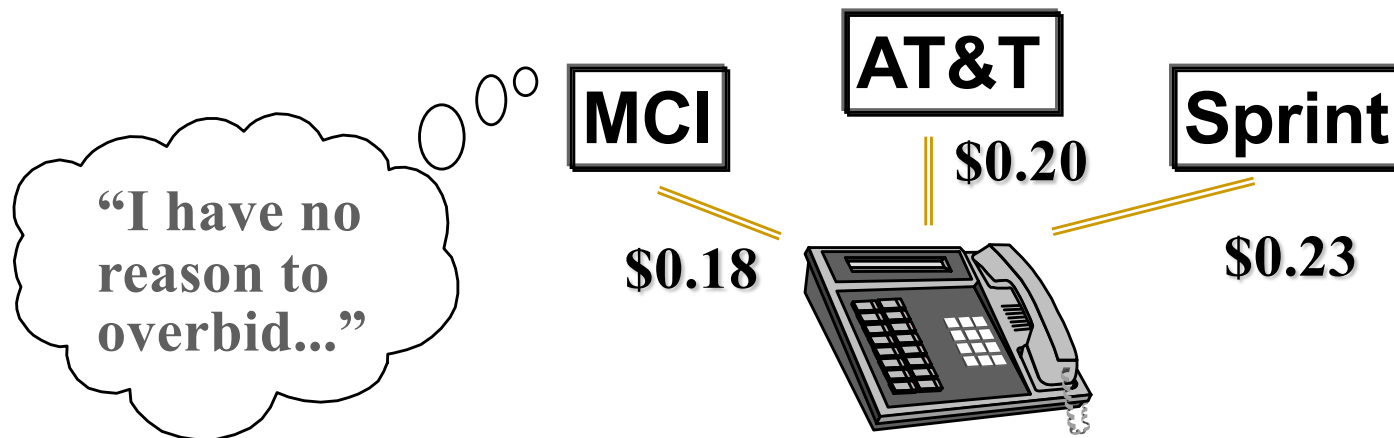
- Phone chooses carrier with lowest bid
- Carrier gets amount of second-best price



Attributes of the Vickrey Mechanism

- ✓ *Distributed*
- ✓ *Symmetric*
- ✓ *Stable*
- ✓ *Simple*
- ✓ *Efficient*

Carriers have *no* incentive to invest effort in strategic behavior



Lies and Collusion

- The various auction protocols are susceptible to lying on the part of the auctioneer, and collusion among bidders, to varying degrees
- All four auctions (English, Dutch, First-Price Sealed Bid, Vickrey) can be manipulated by bidder collusion
- A dishonest auctioneer can exploit the Vickrey auction by lying about the 2nd-highest bid
- *Shills* can be introduced to inflate bidding prices in English auctions

Negotiation

- Auctions are *only* concerned with the allocation of goods: richer techniques for reaching agreements are required
- *Negotiation* is the process of reaching agreements on matters of common interest
- Any negotiation setting will have four components:
 - A negotiation set: possible proposals that agents can make
 - A protocol
 - Strategies, one for each agent, which are private
 - A rule that determines when a deal has been struck and what the agreement deal is
- Negotiation usually proceeds in a series of rounds, with every agent making a proposal at every round

Negotiation in Task-Oriented Domains

Imagine that you have three children, each of whom needs to be delivered to a different school each morning. Your neighbor has four children, and also needs to take them to school. Delivery of each child can be modeled as an indivisible task. You and your neighbor can discuss the situation, and come to an agreement that it is better for both of you (for example, by carrying the other's child to a shared destination, saving him the trip). There is no concern about being able to achieve your task by yourself. The worst that can happen is that you and your neighbor won't come to an agreement about setting up a car pool, in which case you are no worse off than if you were alone. You can only benefit (or do no worse) from your neighbor's tasks. Assume, though, that one of my children and one of my neighbors' children both go to the same school (that is, the cost of carrying out these two deliveries, or two tasks, is the same as the cost of carrying out one of them). It obviously makes sense for both children to be taken together, and only my neighbor or I will need to make the trip to carry out both tasks.

Domain Theory

■ Task Oriented Domains

- ◆ Agents have tasks to achieve
- Task redistribution

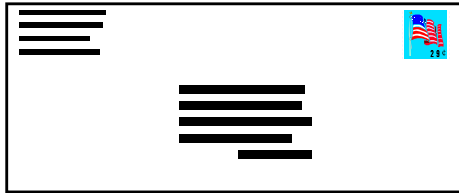
■ State Oriented Domains

- ◆ Goals specify acceptable final states
- ◆ Side effects
- Joint plan and schedules

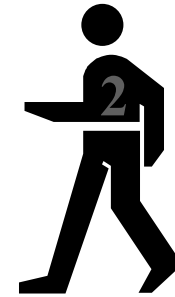
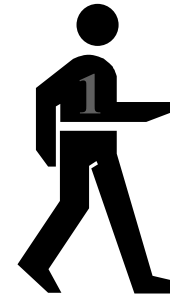
■ Worth Oriented Domains

- ◆ Function rating states' acceptability
- Joint plan, schedules, and goal relaxation

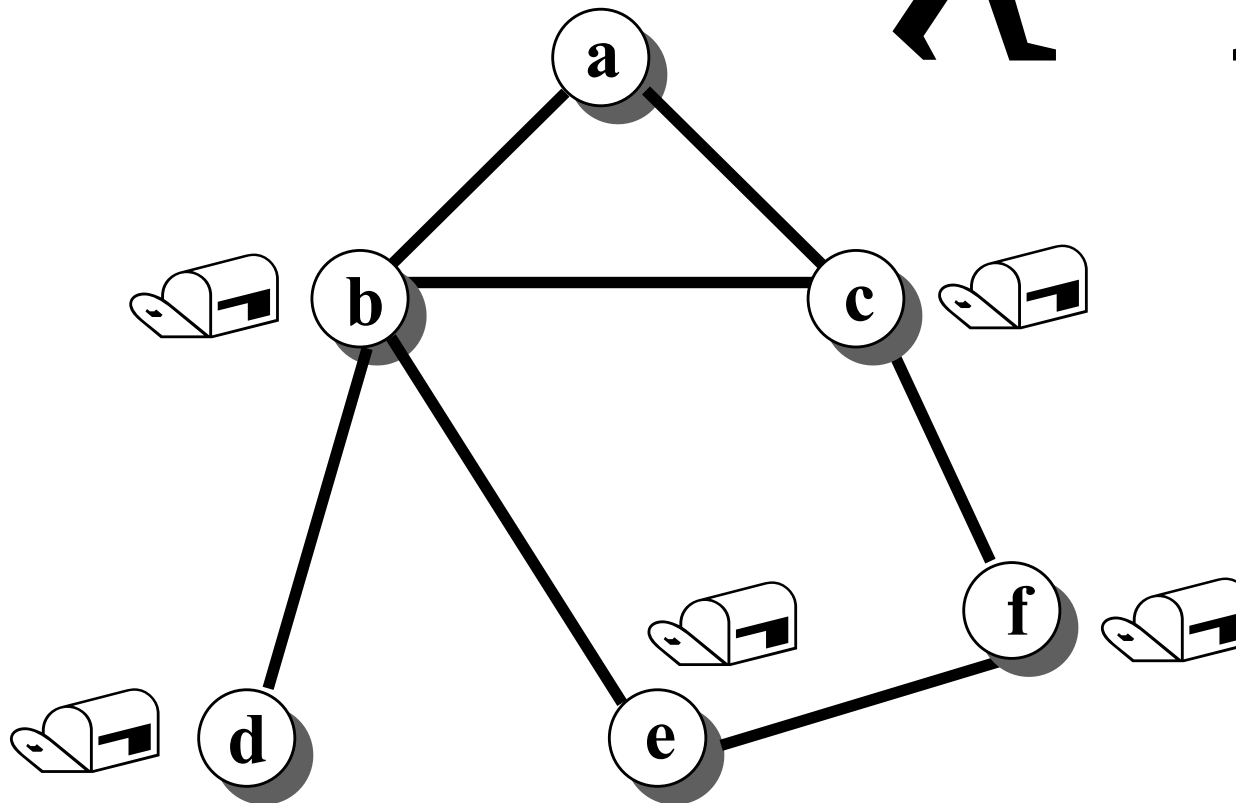
Postmen Domain



Post Office



TOD



TODs Defined

- A TOD is a triple

$$\langle T, Ag, c \rangle$$

where

- T is the (finite) set of all possible tasks
- $Ag = \{1, \dots, n\}$ is the set of participating agents
- $c = \wp(T) \rightarrow \mathbb{U}^+$ defines the *cost* of executing each subset of tasks

- An *encounter* is a collection of tasks

$$\langle T_1, \dots, T_n \rangle$$

where $T_i \subseteq T$ for each $i \in Ag$

Building Blocks

- ✓ Domain
 - A precise definition of what a goal is
 - Agent operations
- Negotiation Protocol
 - A definition of a deal
 - A definition of utility
 - A definition of the conflict deal
- Negotiation Strategy
 - In Equilibrium
 - Incentive-compatible

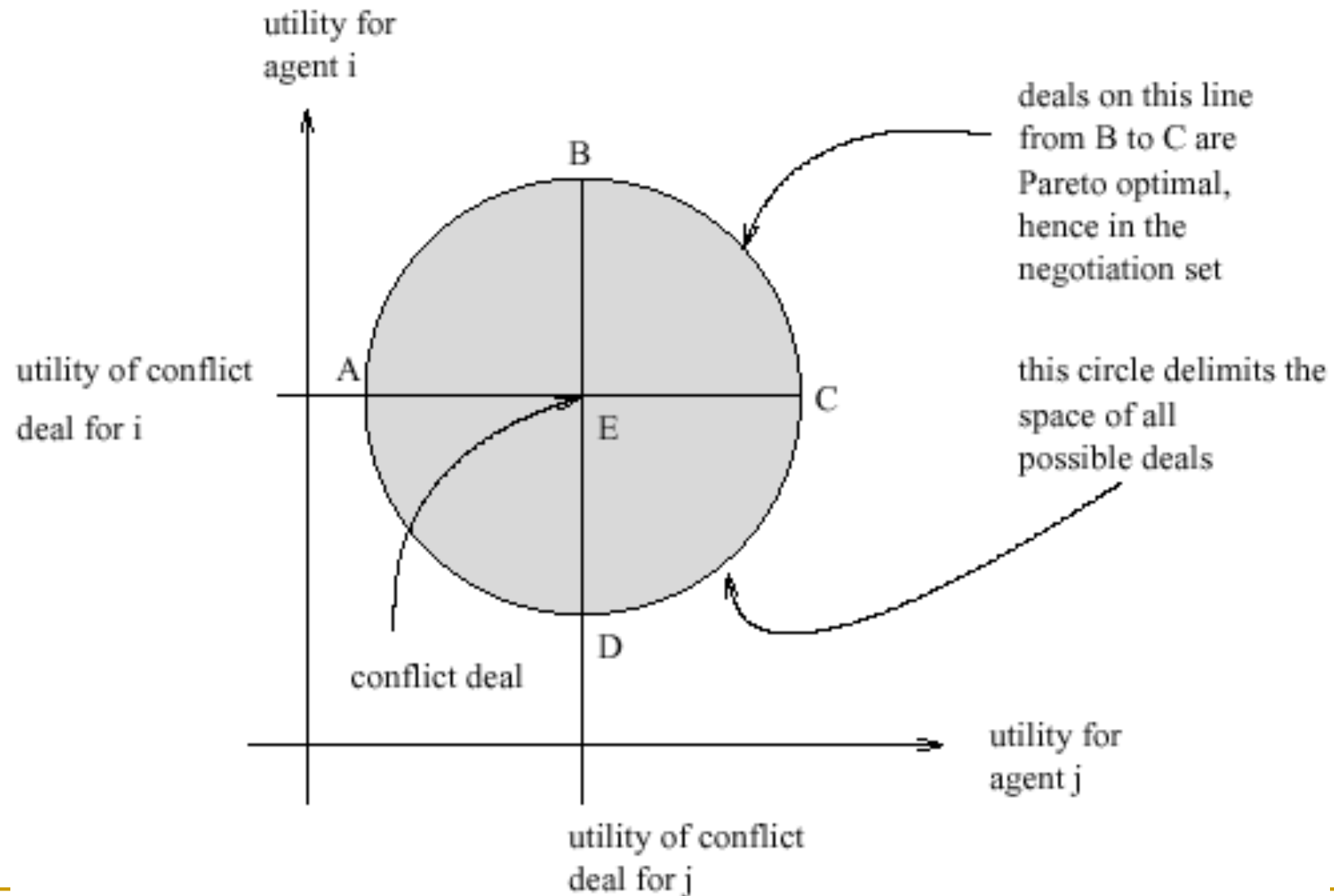
Deals in TODs

- Given encounter $\langle T_1, T_2 \rangle$, a *deal* is an allocation of the tasks $T_1 \cup T_2$ to the agents 1 and 2
- The *cost* to i of deal $\delta = \langle D_1, D_2 \rangle$ is $c(D_i)$, and will be denoted $cost_i(\delta)$
- The *utility* of deal δ to agent i is:
$$utility_i(\delta) = c(T_i) - cost_i(\delta)$$
- The *conflict deal*, Θ , is the deal $\langle T_1, T_2 \rangle$ consisting of the tasks originally allocated. Note that $utility_i(\Theta) = 0$ for all $i \in Ag$
- Deal δ is *individual rational* if it weakly dominates the conflict deal

The Negotiation Set

- The set of deals over which agents negotiate are those that are:
 - individual rational
 - pareto efficient

The Negotiation Set Illustrated



Negotiation Protocols

- Agents use a product-maximizing negotiation protocol (as in Nash bargaining theory)
- It should be a symmetric PMM (product maximizing mechanism)
- Examples: 1-step protocol, monotonic concession protocol...

The Monotonic Concession Protocol

Rules of this protocol are as follows...

- Negotiation proceeds in rounds
- On round 1, agents simultaneously propose a deal from the negotiation set
- Agreement is reached if one agent finds that the deal proposed by the other is at least as good or better than its proposal
- If no agreement is reached, then negotiation proceeds to another round of simultaneous proposals
- In round $u + 1$, no agent is allowed to make a proposal that is less preferred by the other agent than the deal it proposed at time u
- If neither agent makes a concession in some round $u > 0$, then negotiation terminates, with the conflict deal

The Zeuthen Strategy

Three problems:

- What should an agent's first proposal be?
Its most preferred deal
- On any given round, *who should concede?*
The agent least willing to risk conflict
- If an agent concedes, then *how much* should it concede?
Just enough to change the balance of risk

Willingness to Risk Conflict

- Suppose you have conceded a *lot*. Then:
 - Your proposal is now near the conflict deal
 - In case conflict occurs, you are not much worse off
 - You are *more willing* to risk conflict
- An agent will be *more willing* to risk conflict if the difference in utility between its current proposal and the conflict deal is *low*

Risk and willingness to concede

- Risk of agent i at round t :

$$risk_i^t = \frac{\text{utility } i \text{ loses by conceding and accepting } j\text{'s offer}}{\text{utility } i \text{ loses by not conceding and causing conflict}}$$

since $utility(\Theta) = const$ we can set $utility_i(\Theta) = 0$ so that

$$risk_i^t = \begin{cases} 1 & \text{if } utility_i(\delta_i^t) = 0 \\ \frac{utility_i(\delta_i^t) - utility_i(\delta_j^t)}{utility_i(\delta_i^t)} & \text{otherwise} \end{cases}$$

Nash Equilibrium Again...

- The Zeuthen strategy is in Nash equilibrium: under the assumption that one agent is using the strategy the other can do no better than use it himself...
- This is of particular interest to the designer of automated agents. It does away with any need for secrecy on the part of the programmer. An agent's strategy can be publicly known, and no other agent designer can exploit the information by choosing a different strategy. In fact, it is desirable that the strategy be known, to avoid inadvertent conflicts.

Building Blocks

✓ Domain

- ❑ A precise definition of what a goal is
- ❑ Agent operations

✓ Negotiation Protocol

- ❑ A definition of a deal
- ❑ A definition of utility
- ❑ A definition of the conflict deal

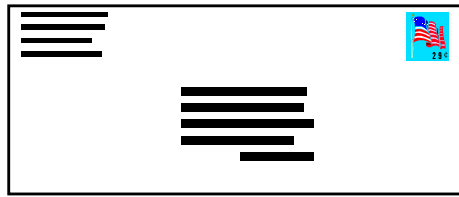
■ Negotiation Strategy

- ❑ In Equilibrium
- ❑ Incentive-compatible

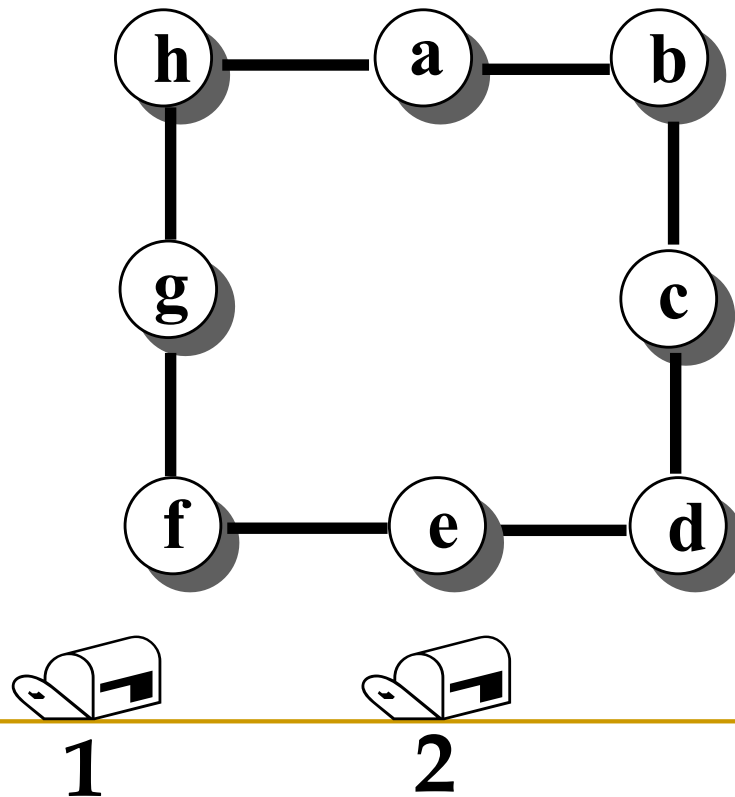
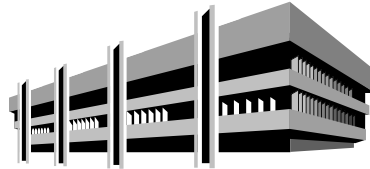
Deception in TODs

- Deception can benefit agents in two ways:
 - *Phantom and Decoy tasks*
Pretending that you have been allocated tasks you have not
 - *Hidden tasks*
Pretending *not* to have been allocated tasks that you have been

Negotiation with Incomplete Information

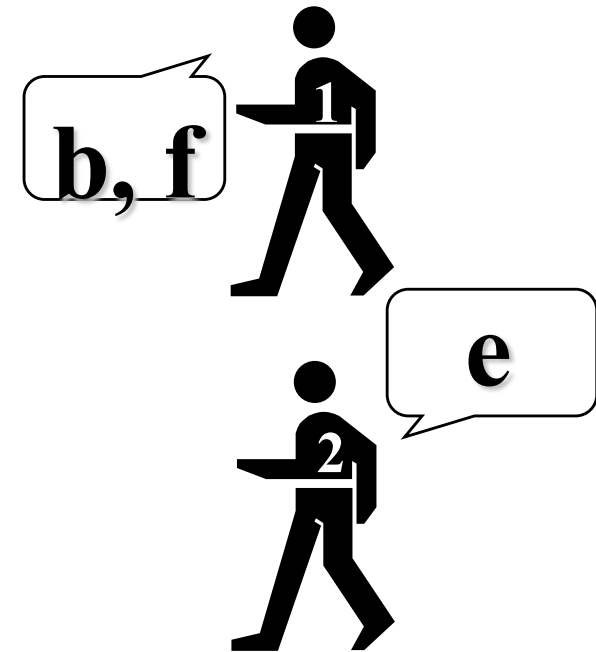
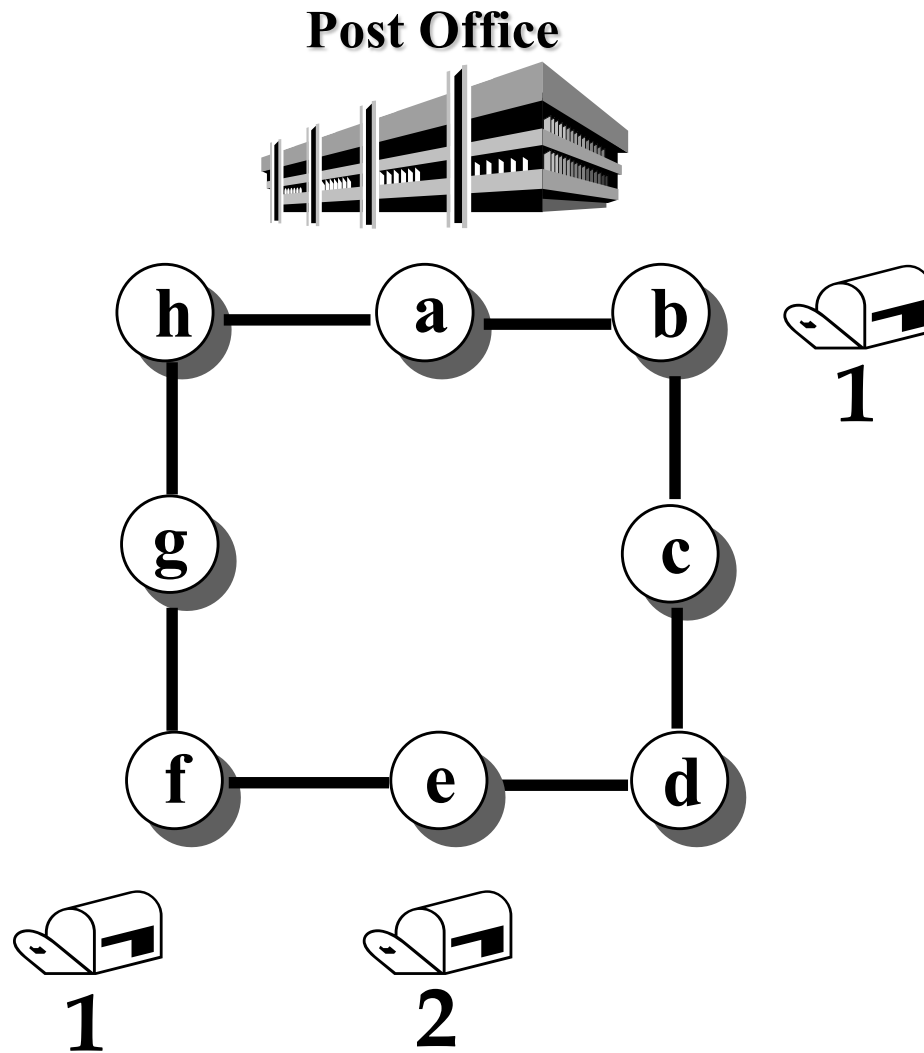


Post Office



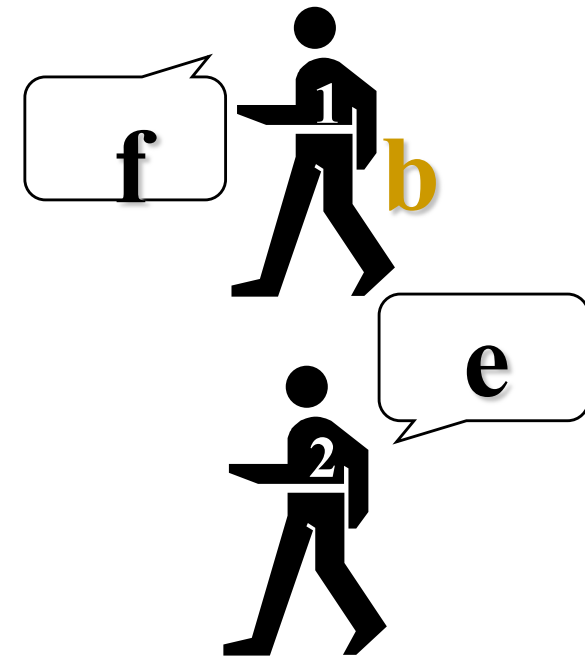
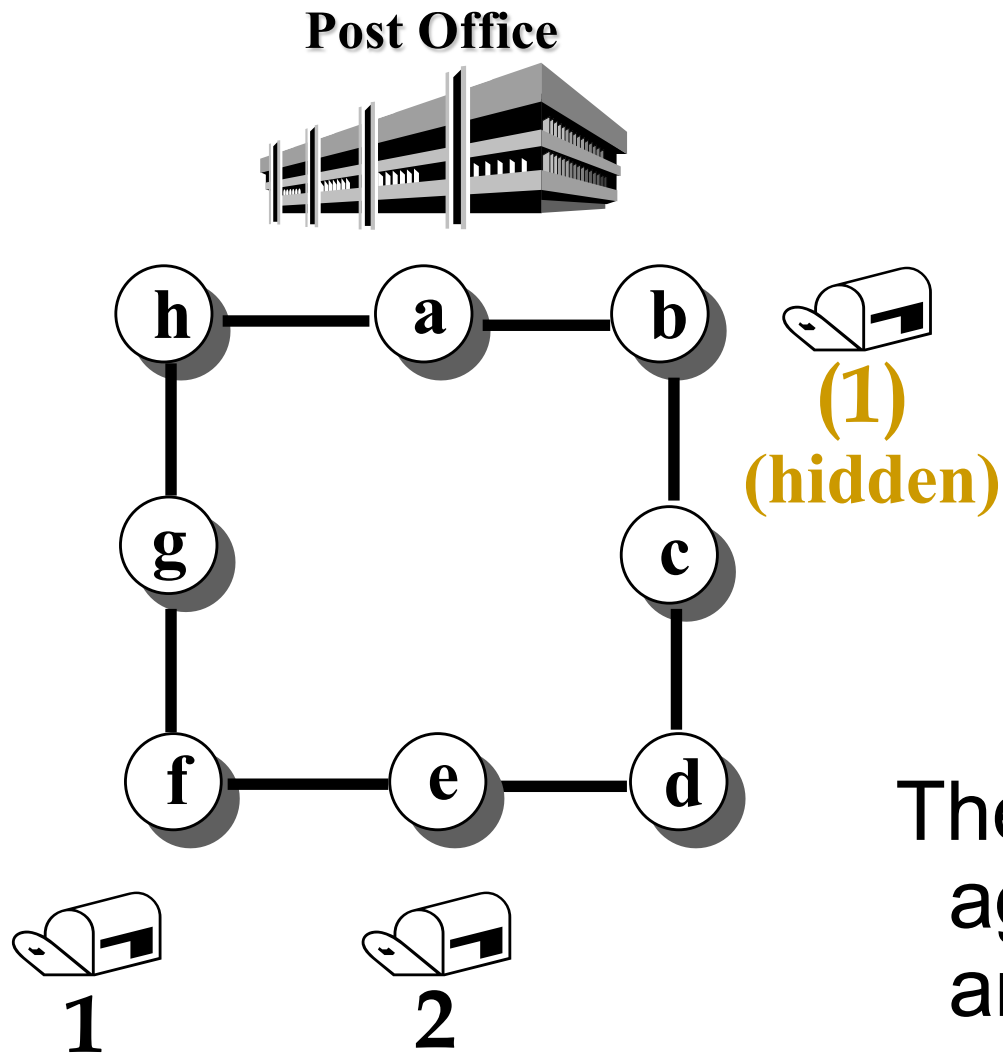
What if the agents don't know each other's letters?

-1 Phase Game: Broadcast Tasks



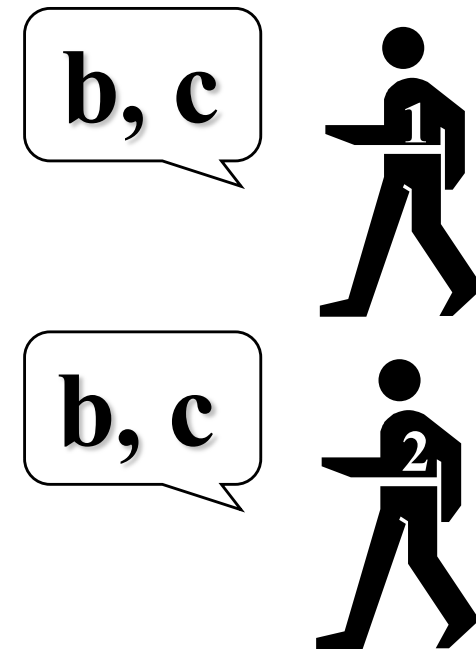
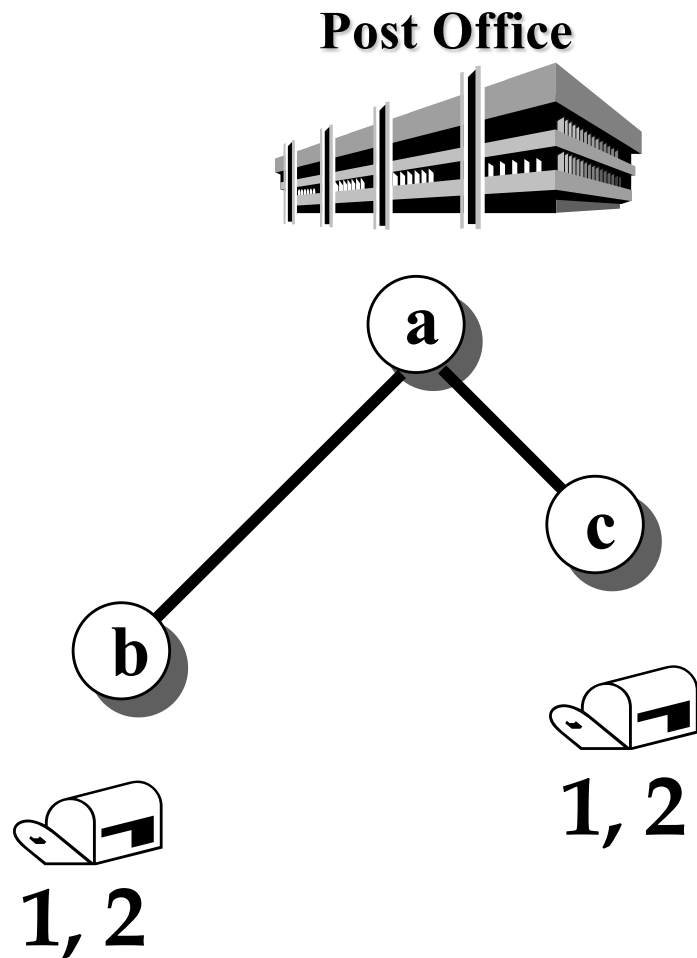
Agents will flip a coin to decide who delivers all the letters

Hiding Letters



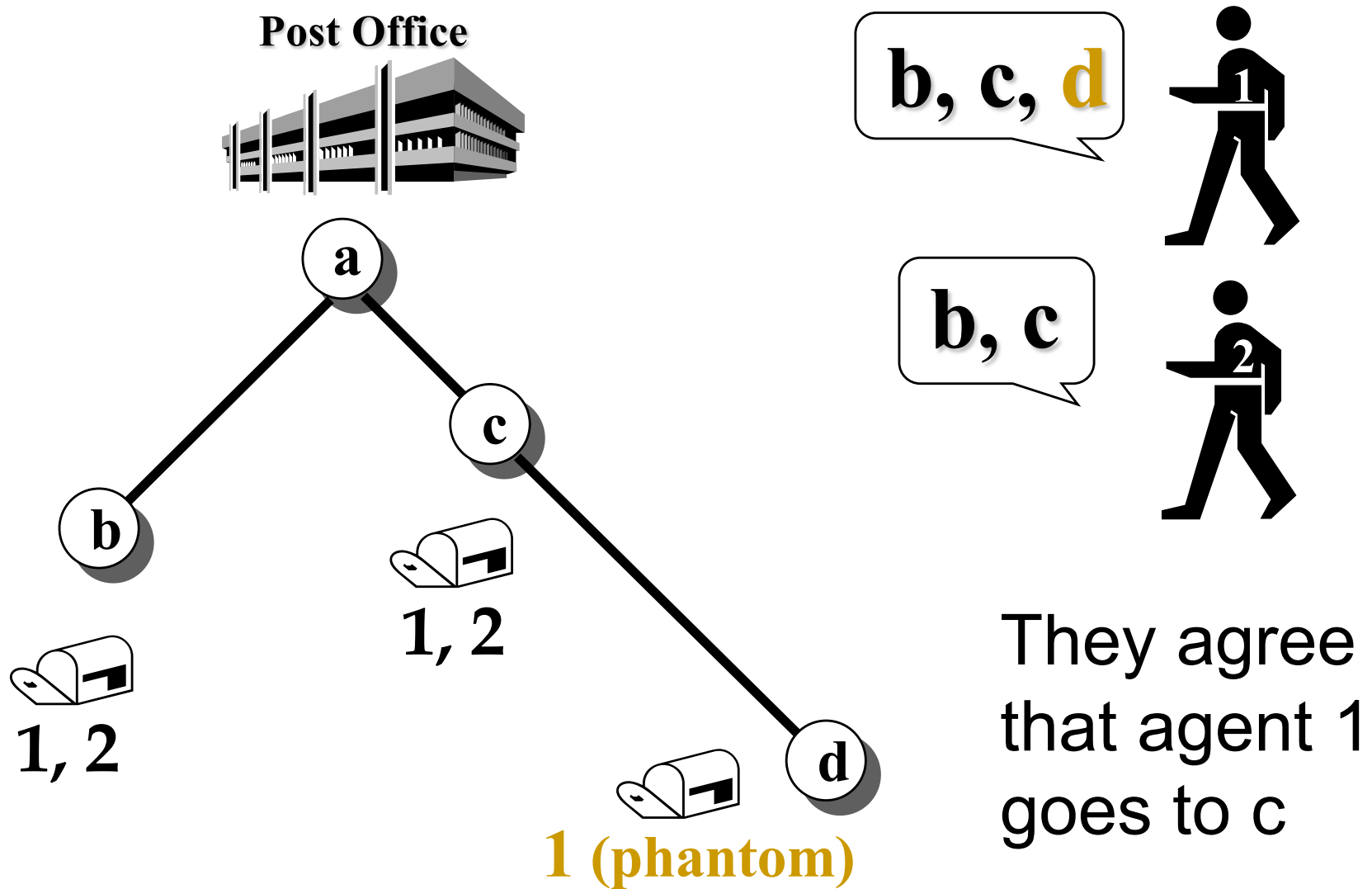
They then agree that
agent 2 delivers to f
and e

Another Possibility for Deception



They will agree to flip a coin to decide who goes to b and who goes to c

Phantom Letter



Worth Oriented Domains

- Instead of distributing tasks: (collaboratively) reach environment state with greatest worth
- Agents try to reach valuable states by negotiating and executing joint plans
- Formally: A WOD is a tuple

$$\langle E, Ag, J, c \rangle$$

- E : Finite set of possible environment states
- $Ag = \{1, \dots, n\}$: Finite set of Agents
- J : set of possible joint plans. A plan (action sequence) that transforms state e_1 into e_2 is written $j: e_1 \rightsquigarrow e_2$
- $c: J \times Ag \rightarrow \mathbb{R}$: Cost function. $c(j, i)$: cost of agent i executing plan j .

Worth Oriented Domains

- An encounter in a WOD is a tuple $\langle e_0, W \rangle$
 - $e_0 \in E$: initial environment state
 - $W: E \times Ag \rightarrow \mathbb{R}$: Worth function. $W(e, i)$ is worth of environment state e for agent i

- If agent operates alone and finds plan $j_{opt}^i \rightarrow$ no negotiation necessary, just execute j_{opt}^i :

$$j_{opt}^i = \operatorname{argmax}_{j \in J: e_0 \rightsquigarrow e} (W(i, e) - C(j, i))$$

- In a MAS: Further improvements possible through joint plans $\rightarrow$ if conflict in best interest states $\rightarrow$ Negotiation

Argumentation

- Argumentation is the process of attempting to convince others of something
- Gilbert (1994) identified 4 modes of argument:
 1. *Logical mode*
“If you accept that *A* and that *A* implies *B*, then you must accept that *B*”
 2. *Emotional mode*
“How would you feel if it happened to you?”
 3. *Visceral mode*
“Cretin!”
 4. *Kisceral mode*
“This is against Christian teaching!”

Logic-based Argumentation

Basic form of logical arguments is as follows:

Database | (*Sentence*, *Grounds*)

where:

- *Database* is a (possibly inconsistent) set of logical formulae
- *Sentence* is a logical formula known as the *conclusion*
- *Grounds* is a set of logical formulae such that:
 1. *Grounds* $\subseteq$ *Database*; and
 2. *Sentence* can be proved from *Grounds*

Attack and Defeat

- Let (ϕ_1, Γ_1) and (ϕ_2, Γ_2) be arguments from some database $\Delta \dots$
Then (ϕ_2, Γ_2) can be defeated (attacked) in one of two ways:
- (ϕ_1, Γ_1) *rebuts* (ϕ_2, Γ_2) if $\phi_1 / \neg \phi_2$
- (ϕ_1, Γ_1) *undercuts* (ϕ_2, Γ_2) if $\phi_1 / \neg \psi_2$ for some $\psi \in \Gamma_2$

A rebuttal or undercut is known as an *attack*

Abstract Argumentation

- Concerned with the overall structure of the argument (rather than internals of arguments)
- Write $x \rightarrow y$
 - “argument x attacks argument y ”
 - “ x is a counterexample of y ”
 - “ x is an attacker of y ”

where we are not actually concerned as to what x, y are

- An *abstract argument system* is a collection of arguments together with a relation “ $\rightarrow$ ” saying what attacks what
- An argument is *out* if it has an undefeated attacker, and *in* if all its attackers are defeated

An Example Abstract Argument System

